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Entropy and Venn Diagrams

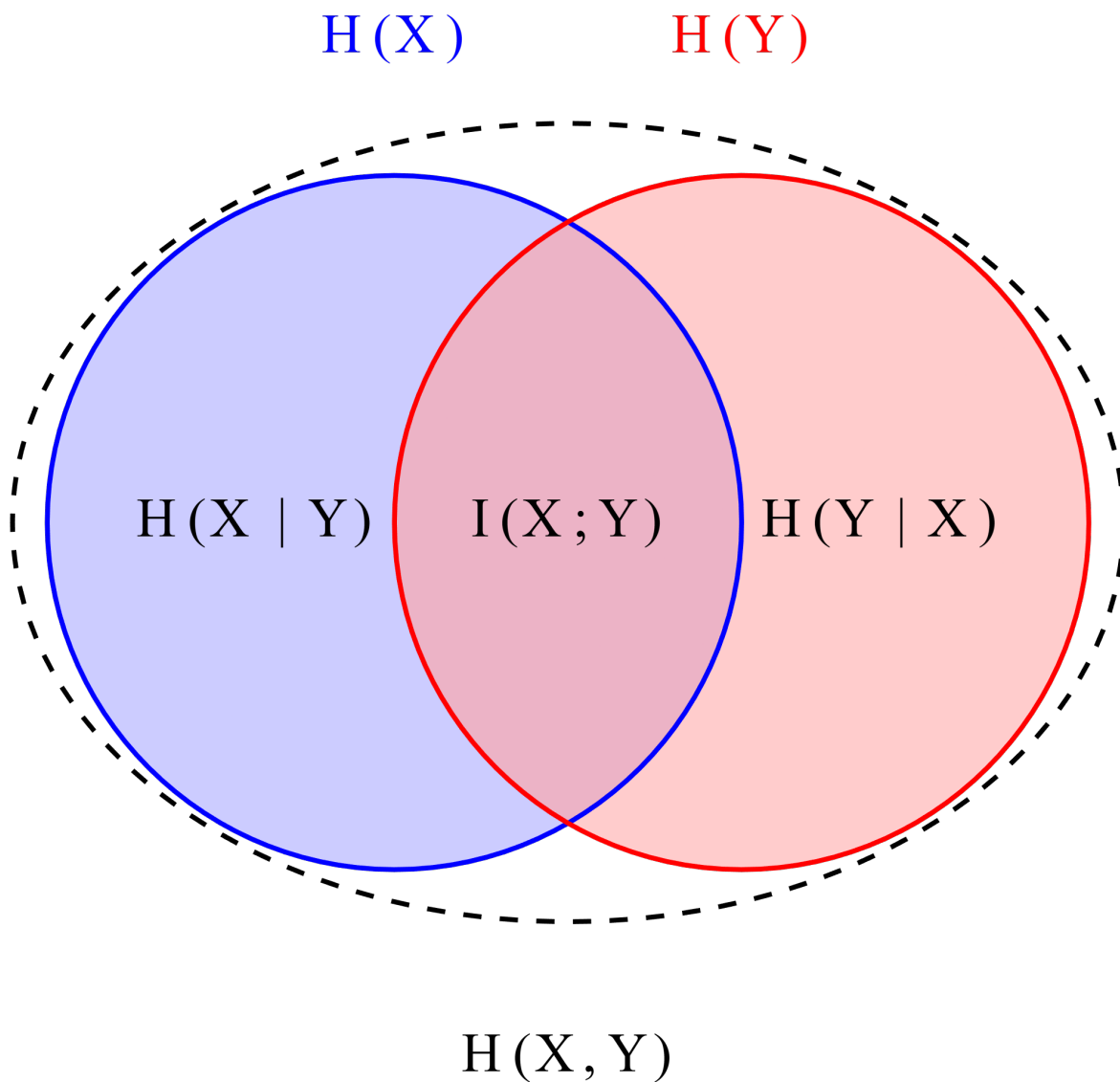
Question

Explain the properties of entropy using examples of conditional entropy $H(X|Y)$ and joint entropy $H(X,Y)$.

Use Venn diagrams.

Introduction to Basic Concepts

Before discussing properties, let's recall the essential definitions illustrated by the following Venn diagram:



Legend:

- $H(X)$: Entropy of X (total uncertainty about X)
- $H(Y)$: Entropy of Y (total uncertainty about Y)
- $H(X, Y)$: Joint entropy (total uncertainty of the pair (X, Y))
- $H(X|Y)$: Conditional entropy (remaining uncertainty about X when Y is known)
- $H(Y|X)$: Conditional entropy (remaining uncertainty about Y when X is known)
- $I(X; Y)$: Mutual information (information shared between X and Y)

Fundamental Properties Illustrated

1. Conditioning Reduces Entropy

Property: $H(X|Y) \leq H(X)$

Intuitive explanation: Knowing Y reduces (or at worst doesn't change) the uncertainty about X .

Illustration: In the diagram, $H(X|Y)$ is the part of circle $H(X)$ that doesn't intersect $H(Y)$. This region is always smaller than or equal to the entire $H(X)$.

Extreme cases:

- If X and Y are independent: $H(X|Y) = H(X)$ (circles are disjoint)
- If Y completely determines X : $H(X|Y) = 0$ (circles overlap completely)

2. Joint Entropy as Union of Uncertainties

Property: $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$

Intuitive explanation: The total uncertainty about the pair (X, Y) equals the uncertainty about X plus the remaining uncertainty about Y once X is known (or vice versa).

Visual formula:

- $H(X, Y)$ = entire area of both circles combined
- $H(X) + H(Y|X)$ = complete $H(X)$ circle + part of $H(Y)$ not covered by X

3. Subadditivity of Joint Entropy

Property: $H(X, Y) \leq H(X) + H(Y)$

Intuitive explanation: The total uncertainty about (X, Y) cannot exceed the sum of individual uncertainties, unless X and Y are independent (in which case it's equal).

Illustration: The area of unification of two circles is always the sum of individual areas. Equality ($H(X, Y) = H(X) + H(Y)$) occurs when circles are disjoint (X and Y independent).

4. Triangular Relationship Between Measures

Property: $H(X, Y) = H(X) + H(Y) - I(X; Y)$

Intuitive explanation: Joint entropy is the sum of individual entropies minus what they share in common (to avoid double-counting shared information).

Set theory analogy: Like in set theory: $|A \cup B| = |A| + |B| - |A \cap B|$

5. Chain of Successive Conditionings

Property: $H(X|Y) \geq H(X|Y, Z)$

Intuitive explanation: The more information we have (by conditioning on more variables), the smaller the remaining uncertainty becomes.

Extended illustration: If we add a third circle Z to the diagram, the region corresponding to $H(X|Y, Z)$ is smaller than $H(X|Y)$.

Concrete Example with Quantitative Diagram

Consider two binary variables X and Y with the following joint distribution:

X	Y	P(X,Y)
0	0	0.4
0	1	0.1
1	0	0.2
1	1	0.3

Let's calculate the values:

- $H(X) \approx 0.971$ bits
- $H(Y) \approx 0.881$ bits
- $H(X, Y) \approx 1.846$ bits
- $H(X|Y) = H(X, Y) - H(Y) \approx 0.965$ bits
- $I(X; Y) = H(X) - H(X|Y) \approx 0.006$ bits (weak dependence)

Observations from this example:

1. $H(X|Y) = 0.965 < H(X) = 0.971$: conditioning slightly reduces entropy
2. $H(X, Y) = 1.846 < H(X) + H(Y) = 1.852$: slight subadditivity
3. $I(X; Y)$ is very small: variables are almost independent

Practical Applications in Data Science

1. Data Compression

- $H(X|Y)$ represents the part of X that cannot be predicted from Y
- In compression with side information, we only code $H(X|Y)$ instead of $H(X)$
- Example: stereo compression where one channel can be partially predicted from the other

2. Machine Learning

- If $H(X|Y)$ is small, then Y is a good predictor of X
- The reduction $H(X) - H(X|Y) = I(X; Y)$ measures the information that Y provides about X
- In feature selection: we choose Y that minimizes $H(X|Y)$

3. Signal Processing

- In a communication system, $H(X|Y)$ represents the remaining uncertainty after reception
- If the channel is noisy, $H(X|Y) > 0$: uncertainty remains
- A perfect channel would have $H(X|Y) = 0$

Summary of Key Properties

Property	Formula	Visual Interpretation	Equality Condition
Conditioning reduces	$H(X Y) \leq H(X)$	Part of H(X) outside intersection	X and Y independent
Joint entropy additive	$H(X, Y) = H(X) + H(Y X)$	Union = first circle + exclusive part of second	Always true
Subadditivity	$H(X, Y) \leq H(X) + H(Y)$	Union sum of circles	X and Y independent
Mutual information	$I(X; Y) = H(X) - H(X Y)$	Intersection of circles	X and Y independent I=0

Pedagogical Conclusion

The properties of conditional and joint entropy can be summarized by a simple geometric intuition:

- $H(X)$ is a circle representing uncertainty about X
- $H(Y)$ is another circle
- $H(X|Y)$ is what remains of circle X when we remove the intersection with Y
- $H(X,Y)$ is the union of both circles
- $I(X;Y)$ is the overlapping area

This visualization explains why:

1. Knowing Y can only help reduce uncertainty about X (never increase it)
2. Uncertainty about (X,Y) is at most the sum of individual uncertainties
3. Shared information “saves” bits in joint coding

These fundamental principles form the basis of data compression, machine learning, and digital communications.